

ECE 573 – Cloud Computing and Cloud Native Systems Summer 2026

Instructor: Professor Jia Wang

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Prerequisites: ECE 242 Digital Computers and Computing

Class Home Page: <https://wngjia.github.io/ece573-web/>

Required Textbook:

- “Cloud Native Go: Building Reliable Services in Unreliable Environments”
Matthew Titmus, O’Reilly Media, 2021, ISBN-13 978-1492076339
- Plus additional research papers

Computer and Network Requirements:

- A recent Windows computer with 4 CPU cores, 16GB memory, and 512GB SSD; or access to a x64 Ubuntu server with 4 CPU cores, 8GB memory, and 100GB storage.
 - We are not able to support ARM-based computers like Apple MacBooks and Raspberry Pi’s, as well as computers that are more than 5 years old.
- Internet access to common code and package repositories like GitHub and Docker Hub.

Course Descriptions: This course introduces students to cloud native systems that build on top of the cloud computing architecture to provide scalable services in dynamic environments. Key topics covered include virtualization and containerization, distributed database systems, communication mechanisms, batch and stream processing, resource management, consensus, security, and system design techniques for scalability, resilience, manageability, and observability. Course projects will provide hand-on experiences on state-of-the-art languages, libraries, and tools with advanced areas.

Grading:

- 5 Homeworks: 2 points each for a total of 10 points.
- 6 Projects: 20 points each for a total of 120 points.
- $A \geq 115$ / $B \geq 100$ / $C \geq 80$.

Homework and Project Policy: Late homeworks and project reports will not be graded. Discussions on homeworks and projects are encouraged, but copying will call for disciplinary action.

Lecture Schedule (tentative):

No.	Week	Topic	Chapters	HW Out	Project Due
1, 2	5/18	Introduction	1,6		
3, 4	5/25	Go Introduction	2,3	HW #1	
5-7	6/1	Virtualization, Containerization		HW #2	PRJ #1
8-10	6/8	Cloud Native Patterns	4,5		PRJ #2
11-13	6/15	Distributed Database Systems		HW #3	
14-16	6/22	Scalability	7		PRJ #3
17-19	6/29	Resource Management		HW #4	
20,21	7/6	Communication Mechanisms	8		PRJ #4
22-24	7/13	Resilience, Manageability, Observability	9-11	HW #5	
25-27	7/20	Batch and Stream Processing			PRJ #5
28-30	7/27	Security, Consensus			
	8/3	No Final Exam			PRJ #6

ADA Statement: Reasonable accommodations will be made for students with documented disabilities. In order to receive accommodations, students must obtain a letter of accommodation from the Center for Disability Resources and make an appointment to speak with me as soon as possible. The Center for Disability Resources is located in the Life Sciences Building, room 218, 312-567-5744 or disabilities@iit.edu.

Sexual Harassment and Discrimination Information: Illinois Tech prohibits all sexual harassment, sexual misconduct, and gender discrimination by any member of our community. This includes harassment among students, staff, or faculty. Sexual harassment of a student by a faculty member or sexual harassment of an employee by a supervisor is particularly serious. Such conduct may easily create an intimidating, hostile, or offensive environment. Illinois Tech encourages anyone experiencing sexual harassment or sexual misconduct to speak with the Office of Title IX Compliance for information on support options and the resolution process. You can report sexual harassment electronically at iit.edu/incidentreport, which may be completed anonymously. You may additionally report by contacting the Title IX Coordinator, Virginia Foster at foster@iit.edu or the Deputy Title IX Coordinator at eespeland@iit.edu. For confidential support, you may reach Illinois Tech's Confidential Advisor at (773) 907-1062. You can also contact a licensed practitioner in Illinois Tech's Student Health and Wellness Center at student.health@iit.edu or (312)567-7550. For a comprehensive list of resources regarding counseling services, medical assistance, legal assistance and visa and immigration services, you can visit the Office of Title IX Compliance website at <https://www.iit.edu/title-ix/resources>.

University Resources for Students' Rights, Responsibilities, and Related Policies:

- The Center for Disability Resources (CDR) Rights and Responsibilities
<https://www.iit.edu/cdr/faqs/rights-and-responsibilities>
- The Family Educational Rights and Privacy Act (FERPA)
<https://www.iit.edu/registrar/students-and-alumni/ferpa>
- Code of Conducts (from Student Handbook)
<https://www.iit.edu/student-affairs/student-handbook/fine-print/code-conduct>
- Policies Procedures and Regulations (from Student Handbook)
<https://www.iit.edu/student-affairs/student-handbook/fine-print/policies-regulations-and-procedures>